Choose a cubic unit for volume

I can... explain why volume is measured in cubic units.

Measure	Unit measures
Length: cm	One direction
Area: cm^2	A 1 cm by 1 cm square
Volume: cm^3	A 1 cm by 1 cm by 1 cm cube

Pack cubes into a solid without gaps or overlaps. Count every cube, including cubes hidden behind others.

A box with 24 one-centimeter cubes has volume 24 cm^3. Counting only visible faces measures the wrong thing.